

Ej Ciclo : falta compras MP + IVA = 2.585.000
 Año comercial para ciclos (360 días)
 Periodo = periodo de las Vtas = ciclo ventas

Ej: Fosforita no necesita amort

$$\text{Ej 1) } R^2 = 0,96 \Rightarrow R = \sqrt{0,96} \Rightarrow R = 0,9797$$

$$\boxed{R \approx 0,980}$$

$$y(7,5) = -7,6 \cdot 7,5 + 120,4 \Rightarrow y = 63,4 \Rightarrow \boxed{5,283 = y \text{ años}}$$

$$y(9,5) = -4,9,5 + 88 \Rightarrow y = 50 \Rightarrow \boxed{4,167 = y \text{ años}}$$

12 meses $\rightarrow$ 1 año
 63,4 meses $\rightarrow$ 5,283

12 meses $\rightarrow$ 1 año
 50 meses $\rightarrow$ 4,167

$$\text{Ej 2) } ROA = \frac{U_{\text{neta}}}{A_T}$$

$$ROE = \frac{U_{\text{neta}}}{PN} = Ra \left(1 + \frac{P}{PN} \right)$$

• Dupont

$$\left\{ \begin{array}{l} Rot A = \frac{V_{\text{tas}}}{A_T} \\ MN = \frac{U_{\text{neta}}}{V_{\text{tas}}} \end{array} \right.$$

$$FM = AC - PC = PNC + PN - ANC$$



$$\text{Ventas} = 3\,900\,000$$

$$\text{CMV} = 1\,649\,700$$

$$\text{Ut Bruta} = 2\,250\,300$$

$$\text{Gs Admin y Com} = 569\,400$$

$$\text{EBITDA} = 1\,680\,900$$

$$\text{Amort y Dep} = 0$$

$$\text{EBIT} = 1\,680\,900$$

$$\text{Gs Financiac} = 700\,000$$

$$\text{EBT} = 980\,900$$

$$\text{Impuestos} = 245\,225$$

$$\text{Ut Neta} = 735\,675$$

$$\text{ROA} = \frac{735\,675}{5\,265\,000}$$

$$\Rightarrow \boxed{\text{ROA} = 0,1397} \quad \text{ROA} = 0,140$$

$$\text{Do} = \frac{\text{Gtos Financ}}{\text{Tint Do}} = \frac{700\,000}{0,34} \Rightarrow \text{Do} = 2\,058\,824$$

$$\text{Prov} = \text{Vtas.} \cdot 0,225 \Rightarrow \text{Prov} = 877\,500$$

$$\text{P} = \text{Do} + \text{Prov} \Rightarrow \text{P} = 2\,936\,324$$

$$\text{PN} = \text{A} - \text{P} \Rightarrow \text{PN} = 2\,328\,676$$

$$\text{ROE} = \frac{735\,675}{2\,328\,676} \Rightarrow \boxed{\text{ROE} = 0,3159}$$

$$\text{Rot A} = \frac{3\,900\,000}{5\,265\,000} \Rightarrow \boxed{\text{Rot A} = 0,7407}$$

$$\text{Ut Bruta} = \text{Vtas.} \cdot 0,577$$

$$\text{Ut Bruta} = 2\,250\,300$$

$$\text{Gs Admin y Com} = 0,146 \cdot 3\,900\,000$$

$$\text{Gs Admin y Com} = 569\,400$$

$$\text{Gs Financ} = 0,225 \cdot \text{Vtas}$$

$$\text{IG} = 0,25 \cdot \text{EBT}$$

$$\text{IG} = 245\,225$$

$$\text{AT} = \text{Vtas.} \cdot 1,35$$

$$\text{AT} = 5\,265\,000$$

$$MN = \frac{735\,675}{3\,900\,000} \Rightarrow 0,1886 \quad \boxed{MN = 0,1886}$$

$$FM = Ac - Pc$$

$$FM = 3\,159\,000 - 1\,906\,912$$

$$\boxed{FM = 1\,252\,088}$$

$$\frac{Ac}{ANc} = 1,5$$

$$Ac = 1,5 (A_T - Ac)$$

$$Ac = \frac{A_T \cdot 1,5}{2,5}$$

$$Ac = 3\,159\,000$$

$$\frac{Do_{cp}}{Do_{ip}} = 1$$

$$Do_{cp} = 1 (Do - Do_{cp})$$

$$Do_{cp} = \frac{Do}{2}$$

$$Do_{cp} = 1\,029\,432$$

$$Pc = Do_{cp} + Prov$$

$$Pc = 1\,906\,912$$

$$Ej\ 3) \quad Ra' = \frac{EBIT(1 - T_{ef})}{A_T}$$

$$EBIT = V_{tas} - CMV - GA_{YC} - Dep \ y \ Amort$$

$$Ra' = \frac{310(1 - 0,3)}{690}$$

$$EBIT = 310$$

$$A_T = 690$$

$$Ra' = 0,31449 \Rightarrow Ra' \approx 0,3145$$

$$FM = A_c - P_c = 255 - 220$$

$$P_c = Prov + D_s Soc + Dep$$

$$FM = 35$$

$$P_c = 220$$

$$A_c = A_T - AF$$

$$A_c = 255$$

$$Wacc = k_e \cdot \frac{PN}{PN + D_o} + k_d \cdot (1 - T_x) \cdot \frac{D_o}{PN + D_o}$$

$$Wacc = 0,27 \cdot \frac{220}{570} + 0,357 \cdot (1 - 0,3) \cdot \frac{350}{570} \quad k_e = 0,27$$

$$T_x = 0,3$$

$$Wacc = 0,257657$$

$$D_o = P - Prov - D_s Soc$$

$$D_o = 350$$

$$Wacc \approx 0,2577$$

$$PN = 220$$

$$k_d = \frac{Gtos \ Finance}{D_o}$$

$$Cap \ inv = NOF + ANC$$

$$Cap \ inv = A_c - Prov - D_s Soc - E_{VO} + AF \quad k_d = 0,357$$

$$Cap \ inv = 555$$

$$Roic = \frac{EBIT(1 - T_x)}{Cap \ inv} = \frac{310(1 - 0,3)}{555} \Rightarrow Roic = 0,391$$

$$EVA = Cap \ inv (Roic - Wacc) = 555 (0,391 - 0,2577)$$

$$EVA = 73,98 \Rightarrow \text{Genera Valor}$$

Ventas - CMU - GA_{YC} - Dep y Amort - Gto Fin - GI = UN

$$\boxed{UN = 120}$$

$$\text{Periodo} = 360/4 = 90$$

$$\text{Ej 5) } PM_{MP} = \frac{\text{saldo medio MP} \cdot 360}{\text{consumo diario MP}} = \frac{635}{2900} \cdot \frac{360}{4}$$

$$PM_{MP} = 0,21896 \cdot \frac{360}{4} \Rightarrow \boxed{PM_{MP} = 19,7}$$

$$PM_{SE} = \frac{(625 + 575)/2}{4125} \cdot \frac{360}{4} \Rightarrow \boxed{PM_{SE} = 13,1}$$

$$PM_{PT} = \frac{(1000 + 970)/2}{4180} \cdot \frac{360}{4} \Rightarrow \boxed{PM_{PT} = 21,2}$$

$$PM_{inu} = PM_{MP} + PM_{SE} + PM_{PT} \Rightarrow PM_{inu} = 54$$

$$PM_{cxc} = \frac{(325 + 375)/2}{5850 \cdot 1,21} \cdot \frac{360}{4} \Rightarrow \boxed{PM_{cxc} = 4,6}$$

$$PM_M = PM_{inu} + PM_{cxc} \Rightarrow PM_M = 58,6$$

$$PM_{PR} = \frac{(475 + 455)/2}{2585} \cdot \frac{360}{4} \Rightarrow PM_{PR} = 16,2$$

$$PM_{EF} = PM_M - PM_{PR} \Rightarrow \boxed{PM_{EF} = 42,4}$$

Ej 6)		\$ std	cons std	\$ real	cons real
MP		70 \$/kg	0,3 kg/u	72 \$/kg	0,25 kg/u
MO		15 \$/h	0,28 h/u	14 \$/h	9000 h

Consumo MP = 12400 kg

compras MP = 5500 kg

$$Q = \frac{\text{Consumo MP}}{\text{cons real MP}} = \frac{12400}{0,25}$$

$$Q = 49600 \text{ uni}$$

$$V_p MP = Q_{\text{comp}} (\$R - \$std) = 5500 \cdot (72 - 70)$$

$$V_p MP = \$11000 \text{ } \boxed{\text{desfav.}}$$

$$V_e MP = \$std (Q_R - Q_{std}) = 70 \cdot (0,25 - 0,3) \cdot 49600$$

$$V_e MP = \$-173600 \text{ } \boxed{\text{fav}}$$

$$V_p MO = Q_R (\$R - \$std) = 9000 (14 - 15)$$

$$V_p MO = \$-9000 \text{ } \boxed{\text{fav}}$$

$$V_e MO = \$std (Q_R - Q_{std} \cdot Q) = 15 (9000 - 0,28 \cdot 49600)$$

$$V_e MO = \$-73320 \text{ } \boxed{\text{fav}}$$

$$\Sigma \text{Variaciones} = -244920 \Rightarrow \boxed{\text{favorable}}$$

$$Ej 4) \Delta U = \sum CM(Q) + V(P) - CV(CV) - CF(CF)$$

$$A: CM = V - CV = 300\,000 - 100\,000$$

$$\left. \begin{array}{l} CM = 200\,000 \\ opQ = 0 \end{array} \right\} \Delta CM = 0$$

$$V = 300\,000$$

$$P_i = \frac{300\,000}{10\,000}$$

$$opP = \frac{P_f - P_i}{P_i} = \frac{29,4 - 30}{30}$$

$$P_i = 30$$

$$opP = -0,02$$

$$\Delta P_f = 30 \cdot 0,02 = 0,6$$

$$V(P) = 300\,000 \cdot (-0,02)$$

$$P_f = 30 - 0,6 = 29,4$$

$$\Delta V(P) = -6000$$

$$\left. \begin{array}{l} CV = 100\,000 \\ opCV = 0 \end{array} \right\} \Delta CV = 0$$

$$\left. \begin{array}{l} CF = 3000 \\ opCF = 1 \end{array} \right\} \Delta CF = 3000$$

$$P_i \cdot e = Q_i$$

$$\Delta U = 0 = 200\,000 \cdot e \cdot (-0,02) + 300\,000 \cdot (-0,02) - 100\,000 \cdot 0$$

$$0 = 200\,000 \cdot e \cdot (-0,02) + 300\,000 \cdot (-0,02)$$

$$e = \frac{300\,000 \cdot (-0,02)}{200\,000 \cdot (-0,02)} \Rightarrow \boxed{e = 1,5}$$

$$OpQ = 1,5 \cdot (-0,02) \Rightarrow OpQ = (-0,03)$$

$$\Delta CM = 200\,000 \cdot (-0,03) = -6000$$

$$\Delta U_A = -6000 - 6000 - 3000 \Rightarrow \Delta U_A = -15000$$

B:

$$CM = 875000 - 387000 = 488000$$

$$opQ = e.op \Rightarrow opQ = 4 \cdot \frac{85,75 - 87,5}{87,5}$$

$$opQ = -0,08$$

$$P_{Vi} = \frac{875000}{10000}$$

$$P_{Vi} = 87,5$$

$$P_{Vf} = 87,5 \cdot 0,98$$

$$P_{Vf} = 85,75$$

$$\Delta CM = CM \cdot opQ = 488000(-0,08)$$

$$\Delta CM = -39040$$

$$\left. \begin{array}{l} V = 875000 \\ opP = -0,02 \end{array} \right\} \Delta V = -17500$$

$$\left. \begin{array}{l} CV = 387000 \\ opCV = 0 \end{array} \right\} \Delta CV = 0$$

$$\left. \begin{array}{l} CF = 4000 \\ opCF = 1 \end{array} \right\} \Delta CF = 4000$$

$$\Delta U_B = -39040 - 17500 - 4000 \Rightarrow \boxed{\Delta U_B = -60540}$$

$$\Delta U_T = \Delta U_A + \Delta U_B = -15000 - 60540 \Rightarrow \Delta U_T = -75540$$

$$U - \Delta U_T = 100000 - 75540 \Rightarrow \boxed{U = 24460}$$